

* Determination of Hardness of water

- 1) O. Hanner's method
- 2) Soap titration method
- 3) EDTA method

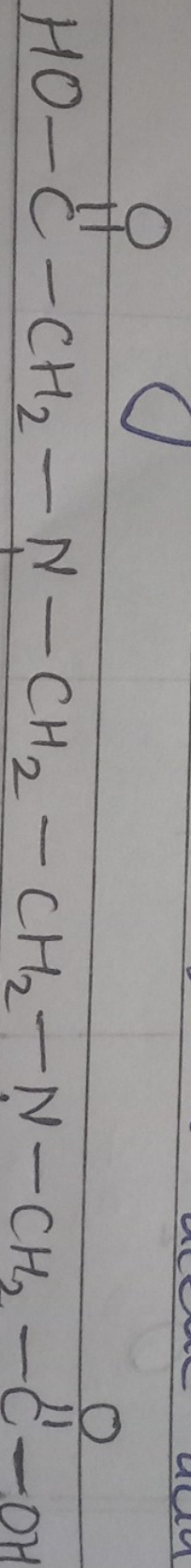
EDTA method for determination of Hardness

OR

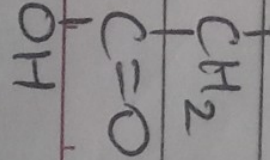
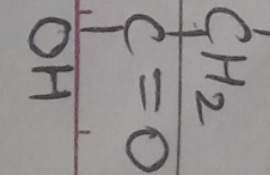
It is also known as complexometric method or titration because EDTA and EBT complexing agents forms complexes with hardness causing Ca^{++} and Mg^{++} salts present in hard water.

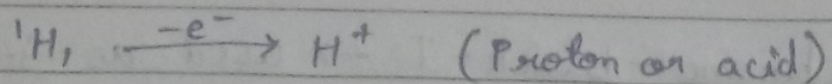
Principle

1) EDTA $\rightarrow$ Ethylene diamine tetra acetic acid



complexing agent

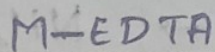




2) It is tetra Protic acid and represented as H_4Y .

3) It's tetra protic form is less soluble in water. Therefore it's disodium dihydrated form is used $\text{Na}_2\text{H}_2\text{Y}$.

Imp 4) EDTA forms stable complex with metal ions or hardness causing ions in the ratio of 1:1 irrespective of the valencies of metal ions.



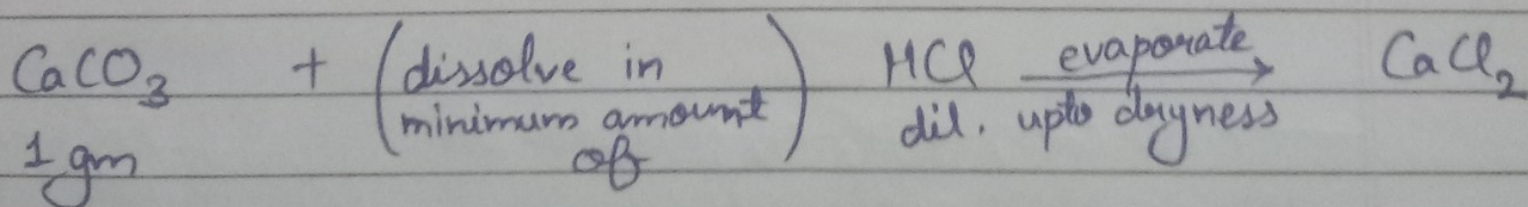
Ex $\Rightarrow$ If 45 ml of EDTA = 45 mg of CaCO_3 eq hardness used for titration

Thus the amount of EDTA used for titration can be taken as the amount of eq. hardness present in water sample.

Chemical Used

1) Standard hard water (SHW)

$\rightarrow$ It's concentration or strength is known.

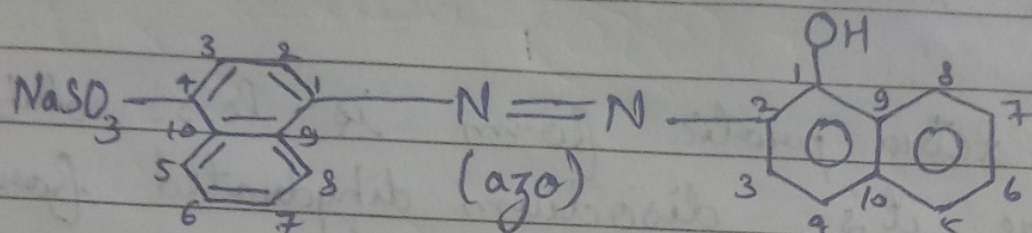


Now make upto volume 1 litre with distil water.

1 lit. SHW = 1g CaCO_3 eq.

1 ml SHW = 1mg of CaCO_3 eq.

2) EBT or Eriochrome-black-T $\rightarrow$ used as indicator.



IUPAC Name is

$\Rightarrow$ Sodium - I (1-hydroxy-2-naphthyl-azo) - 6-nitro-2-naphthol-4-sulphonate

3) Buffer Solution $\rightarrow \text{NH}_4\text{Cl} + \text{NH}_4\text{OH}$
 $\rightarrow$ buffer is used to maintain PH at 10

Why $\rightarrow$

(1) Because at $\text{PH}=10$ EDTA forms stable metal complex.

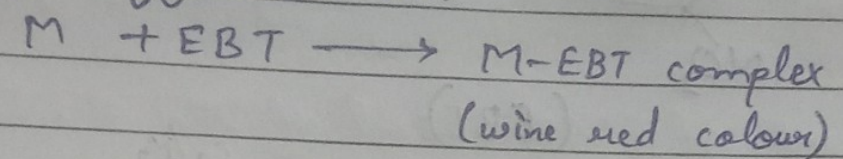
(2) We get distinct or sharp end point at this PH.

4) EDTA solution $\rightarrow$ used for titration.

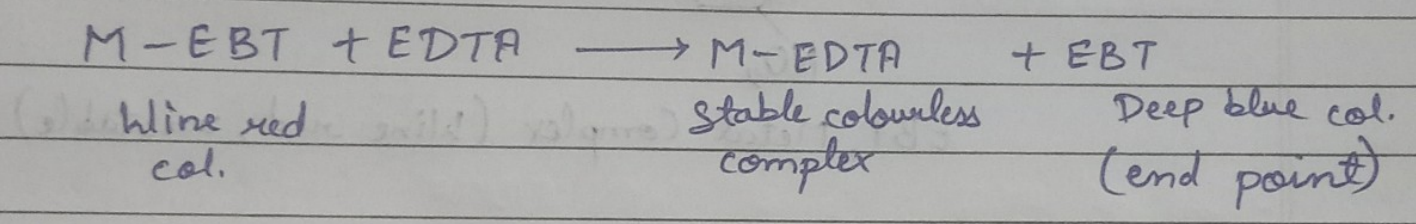
5) Hard water sample $\rightarrow$ whose strength is to be calculated.

Procedure $\Rightarrow$

- 1) 25 ml of SHW/HWS + 1-2 drops of EBT $\rightarrow$ wine red colour solution
in a conical flask
+ 10 ml of buffer solution

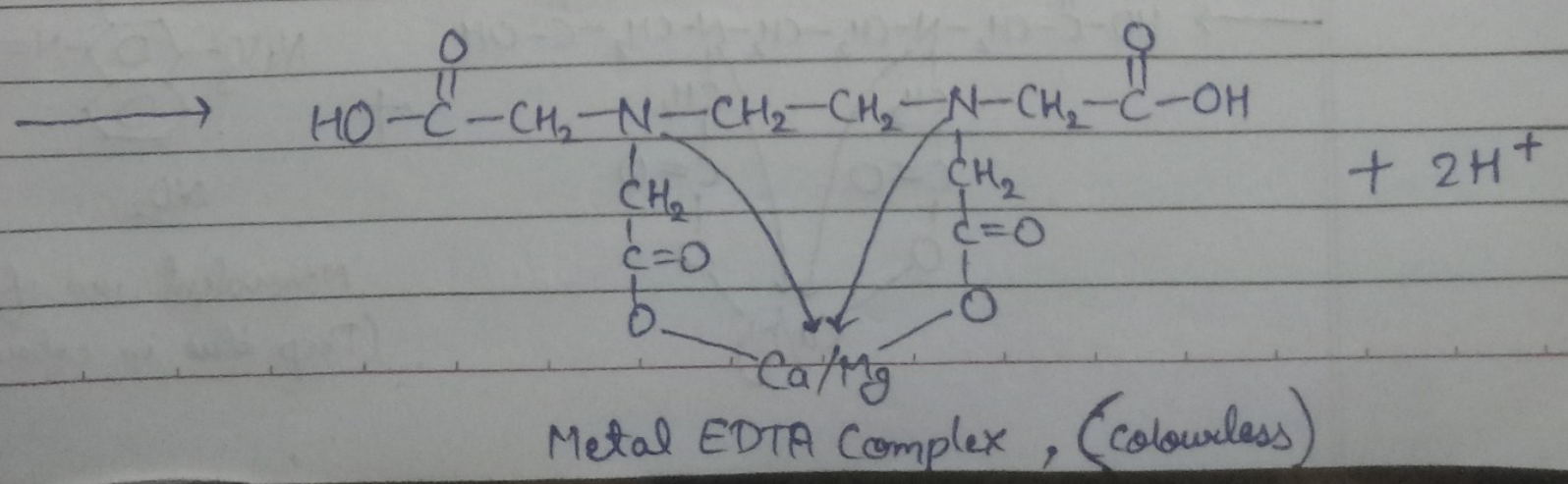
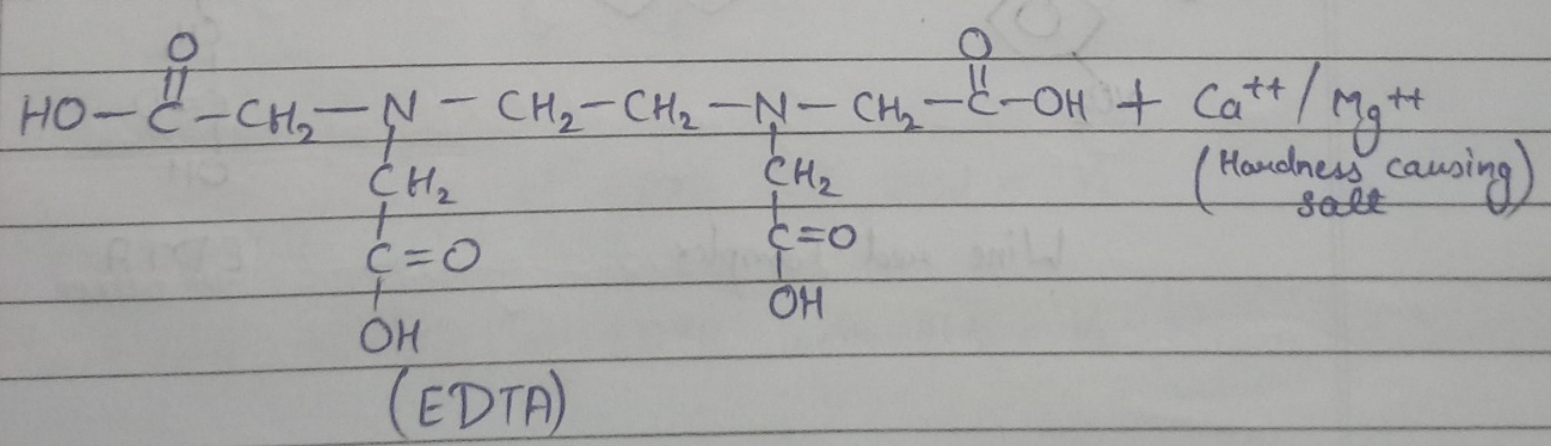


- 2) Add EDTA from burette till wine red sol. converted into deep blue $\rightarrow$ end point of reaction.

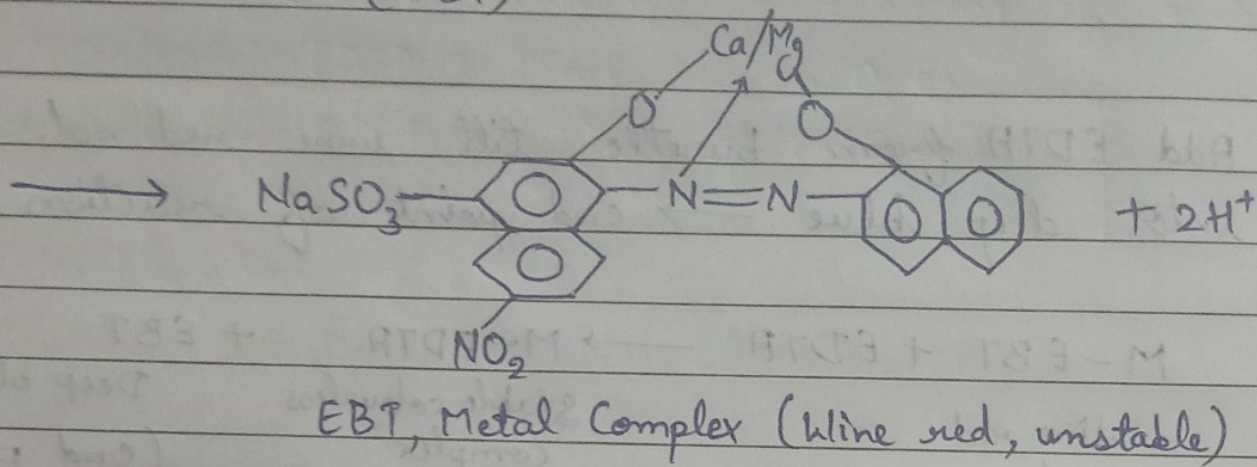
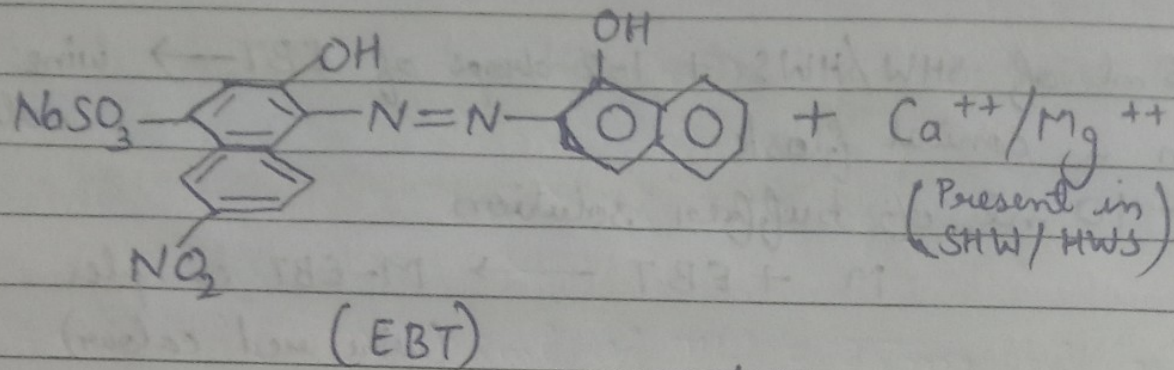


Reactions involved

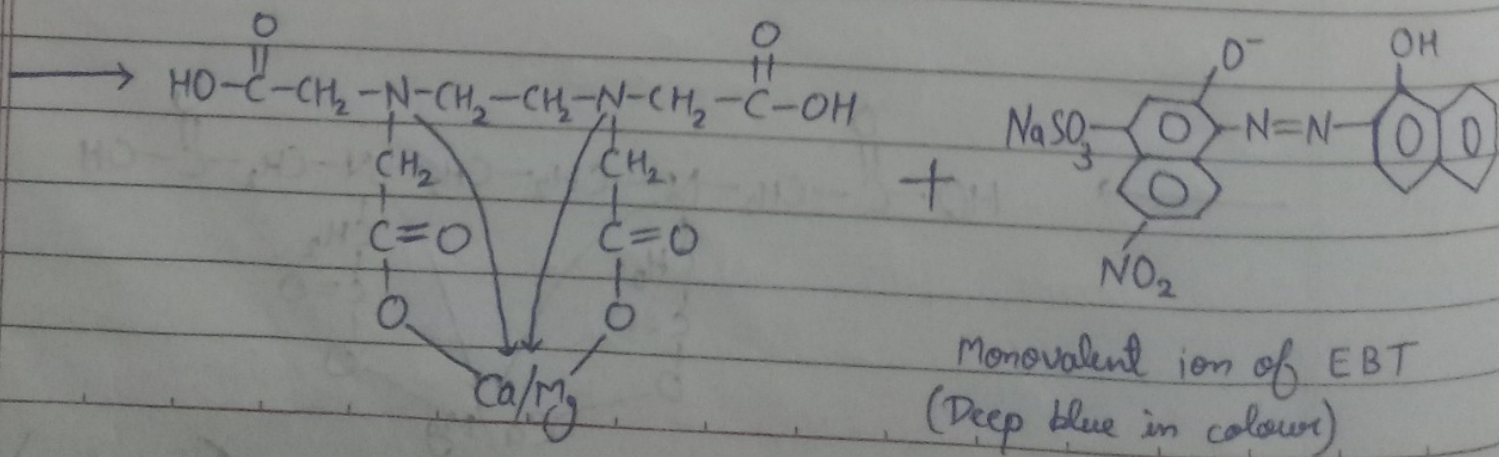
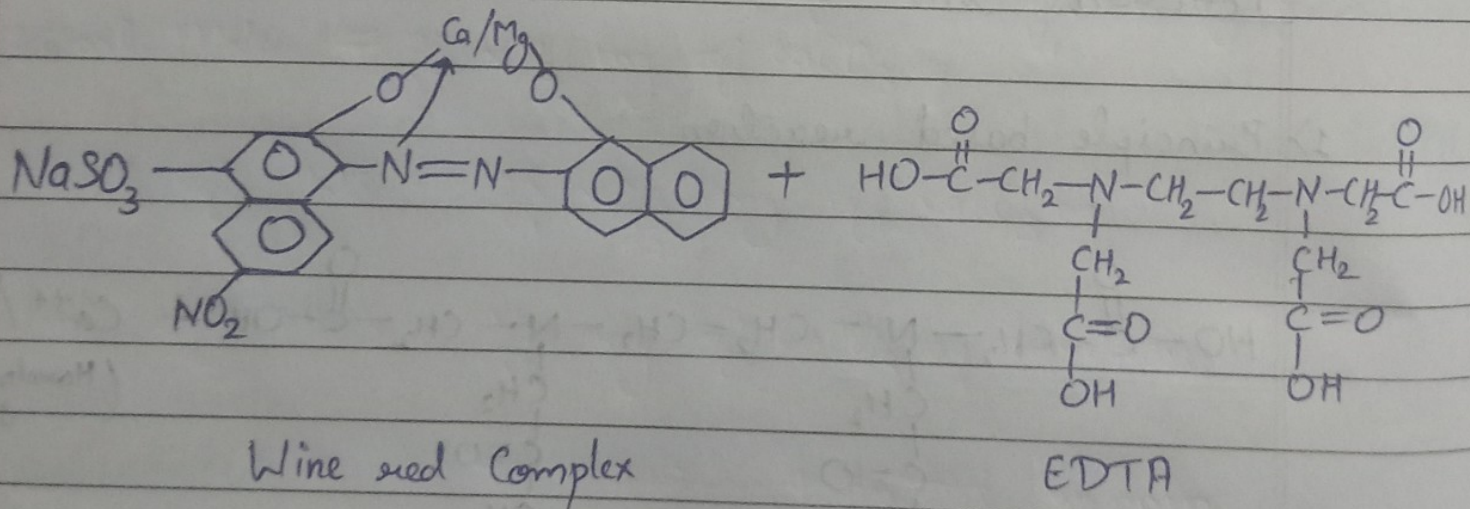
- 1) Principle based reaction $\rightarrow$



2) Reaction between EBT and metal ion or Step I reaction



3) Reaction of wine red complex with EDTA or step II reaction



100 ml of HWS



Boil it till its 80% volume reduced



Due to boiling $\text{Ca}(\text{HCO}_3)_2 / \text{Mg}(\text{HCO}_3)_2$ converts into $\text{CaCO}_3 / \text{Mg}(\text{OH})_2$ ppt



Filter it



Collect filtrate



Wash ppt with distilled water



Make up volume of filtrate as its initial volume.

This HWS having only permanent hardness. $\text{HWS}_{(\text{PH})}$

Consumed volume of EDTA with SHW = V_1

Consumed volume of EDTA with HWS = V_2

Consumed volume of EDTA with $\text{HWS}_{(\text{PH})}$ = V_3

Determination of Permanent and Temporary hardness of water by EDTA Method.

100 ml of HWS

↓

Boil it till its 80% volume reduced

↓

Due to boiling $\text{Ca}(\text{HCO}_3)_2 / \text{Mg}(\text{HCO}_3)_2$ converts into $\text{CaCO}_3 / \text{Mg}(\text{OH})_2$ ppt

↓

Filter it

↓

Collect filtrate

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Wash ppt with distilled water

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Make up volume of filtrate as its initial volume.

This HWS having only permanent hardness. $\text{HWS}_{(\text{PH})}$

Consumed volume of EDTA with SHW = V_1

Consumed volume of EDTA with HWS = V_2

Consumed volume of EDTA with $\text{HWS}_{(\text{PH})}$ = V_3

Calculation -

1 lit of SHW = 1g of CaCO_3 eq. hardness

1000 ml of SHW = 1000mg of CaCO_3 eq

1 ml of SHW = 1mg of CaCO_3 eq.

V_1 ml of EDTA = 25ml of SHW = 25mg of CaCO_3

V_1 ml of EDTA = 25mg of CaCO_3 eq

1 ml of EDTA = $\frac{25}{V_1}$ mg of CaCO_3 eq

Standardization of EDTA

25ml of HWS = V_2 ml of EDTA

when 1ml of EDTA = $\frac{25}{V_1}$ mg of CaCO_3 eq

25 ml HWS = V_1 V_2 ml of EDTA = $\frac{25}{V_1} \times V_2$ mg of CaCO_3 eq

25 ml HWS = $\frac{25}{V_1} \times V_2$ mg of CaCO_3 eq

1ml of HWS = $\frac{25}{V_1} \times \frac{V_2}{25}$ mg of CaCO_3 eq

1ml of HWS = $\frac{V_2}{V_1}$ mg of CaCO_3 eq hardness

= $x \times 1000 = y$ mg/lit or ppm

Que →)

Sol →)